Indian Institute of Engineering Science and Technology, Shibpur
B.Tech. (CST) 3^{rd} Semester Mid-Term Examination, September 2023
Discrete Structures (CS - 2101)

Time: 2 hours **Full Marks: 30**

Answer all questions.
Write all parts of the same question together.

1. (a) Name one of the common fallacies that arise due to the invalid use of the rules of inference.
   (b) Let R be the relation on the set of people consisting of pairs (a, b), where a is a parent of b. Let S be the relation on the set of people consisting of pairs (a, b), where a and b are siblings (brothers or sisters). What are $S \circ R$ and $R \circ S$?
   (c) Let R be the relation on the set $\{0, 1, 2, 3\}$ containing the ordered pairs (0, 1), (1, 1), (1, 2), (2, 0), (2, 2), and (3, 0). Find the (i) reflexive closure of R; (ii) symmetric closure of R
   (d) What is the mathematical logic representation for the statement "Some real numbers are rational."?
   (e) Write the converse, contrapositive, and inverse of the following conditional statement: "I go to the beach whenever it is a sunny summer day."
   (f) Let $A = \{1, 2, 3\}$ be a set. Which of the following relations are functions?
   (i)$\{(1, 2), (2, 3), (1, 3)\}$; (ii) $\{(1, 2), (2, 2), (3, 2)\}$; (iii) $\{(1, 2), (2, 1), (3, 3)\}$; (iv)$\{(1, 2), (2, 3)\}$
   (g) Are these system specifications consistent? "Whenever the system software is being upgraded, users cannot access the file system. If users can access the file system, then they can save new files. If users cannot save new files, then the system software is not being upgraded."
   (h) Construct a compound proposition that asserts that every cell of a 9×9 Sudoku puzzle contains at least one number.

   $[1 + 2 + 2 + 1 + 3 + 2 + 2 + 3 = 16]$

2. Let $Q(x, y)$ be the statement "student x has been a contestant on quiz show y." Express each of these sentences in terms of $Q(x, y)$, quantifiers, and logical connectives, where the domain for x consists of all students at your school and for y consists of all quiz shows on television.
   (a) There is a student at your school who has been a contestant on a television quiz show.
   (b) No student at your school has ever been a contestant on a television quiz show.
   (c) There is a student at your school who has been a contestant on "Jeopardy" and on "Wheel of Fortune."
   (d) Every television quiz show has had a student from your school as a contestant.
   (e) At least two students from your school have been contestants on "Jeopardy."

   $[1 \times 5 = 5]$

3. Use rules of inference to show that the hypotheses "A student in this class has not read the book," "Everyone in this class passed the first exam," imply the conclusion "Someone who passed the first exam has not read the book." [4]

4. Determine whether the relation R represented by the matrix M_R is reflexive, irreflexive, symmetric, anti-symmetric, and/or transitive [give brief justifications].

$$M_R = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

   $[1 + 1 + 1 + 1 + 1 = 5]$